

Absolute-Value and Piecewise Functions — Evaluating Answer Key

Algebra 1

Quick Answers

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|------------|--|
| 1. 6 | 6. 2, 0, 3 |
| 2. 5, 2, 6 | 7. -1, 7 |
| 3. 5 | 8. 11.5, 5 |
| 4. -5, 1 | 9. — |
| 5. 13 | 10. $x = 4$, $y = 13$, $x = -1$, $y = -2$ |
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Curriculum

Level: Algebra 1

HSA-CED.A.1–A.4 – *problems 5, 8*

HSA-REI.C.6, HSA-REI.C.7 – *problem 10*

HSF-IF.A–HSF-IF.C – *problems 1, 2, 3, 4, 6, 7, 9*

Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

4. *If they got 7: used the top branch at the boundary, where the condition is x less than 2*
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What this sheet is really testing: not arithmetic, but the habit of *deciding before substituting*. An absolute-value rule asks the student to finish the inside first; a piecewise rule asks which condition the input satisfies. A student who substitutes first and reads the rule afterwards gets problems 4, 6 and 9 wrong in exactly the same way. When grading, look at the first written line: it should name a branch or show the inside of the bars, not a final number.

1. Evaluating an absolute-value rule at one input.

Substitute $x = -1$ into $f(x) = |x - 3| + 2$ and finish the inside of the bars before anything else:

$$f(-1) = |-1 - 3| + 2 = |-4| + 2 = 4 + 2$$

So $f(-1) = \boxed{6}$.

Why the order matters: the bars are grouping symbols. A student who takes “absolute value of -1 ” first gets $1 - 3 + 2 = 0$, which is not on the V at all.

2. A table of values for the same rule.

Each cell is the same three-step routine: subtract, take the absolute value, add 2.

$$f(0) = |0 - 3| + 2 = 3 + 2 \quad f(3) = |3 - 3| + 2 = 0 + 2 \quad f(7) = |7 - 3| + 2 = 4 + 2$$

x	0	3	7
$f(x)$	5	2	6

Reading the table against the graph: the middle column is the lowest output on the whole table, and it happens at the one input that makes the inside of the bars zero. That is the vertex of the V. Notice also that $x = 0$ and $x = 7$ are not the same distance from 3, so their outputs are different — symmetry only pairs inputs that are equally far from the vertex.

3. A coefficient outside the bars.

The bars close before the 2 does its work, so evaluate inside, take the absolute value, then multiply, then subtract:

$$g(-9) = 2|-9 + 4| - 5 = 2|-5| - 5 = 2(5) - 5 = 10 - 5$$

So $g(-9) = \boxed{5}$.

Watch for: dropping the bars and computing $2(-9 + 4) - 5 = -15$. The bars are not decoration — they are why this function never dips below -5 .

4. Choosing a branch, including at the boundary.

(a) Test the input against the conditions first: is $-2 < 2$? Yes, so the *top* branch, $3x + 1$, is the one that applies.

$$p(-2) = 3(-2) + 1 = -6 + 1$$

So $p(-2) = \boxed{-5}$.

(b) Is $2 < 2$? No. The second condition is $x \geq 2$, and $2 \geq 2$ is true, so the *bottom* branch, $x^2 - 3$, applies.

$$p(2) = 2^2 - 3 = 4 - 3$$

So $p(2) = \boxed{1}$.

Watch for: the boundary. Using the top branch at $x = 2$ gives $3(2) + 1 = 7$, a plausible-looking wrong answer. The $\geq$ sign is what decides it, and it is attached to the second branch.

5. Reading a piecewise model in context.

Four hours is more than 1 and no more than 5, so the middle branch of C applies — the branch that charges the flat first-hour rate plus a per-hour rate after that:

$$C(4) = 4 + 3(4 - 1) = 4 + 3(3) = 4 + 9$$

The driver pays $\boxed{13}$ dollars.

What the pieces mean: the 4 is the price of the first hour no matter how short the stay, the 3 is the price of each additional hour, and the $t - 1$ counts only the hours *after* the first. A student who writes $3(4)$ has charged the customer for a first hour twice.

6. Three branches, one of them a boundary case.

Test each input against the conditions in order.

$-3 < -1$, so the first branch applies: $q(-3) = -(-3) - 1 = 3 - 1 = \boxed{2}$.

$-1 \leq 0 < 1$, so the middle branch applies: $q(0) = 0^2 = \boxed{0}$.

$1 \geq 1$, so the third branch applies: $q(1) = 4 - 1 = \boxed{3}$.

The boundary input is $x = 1$. The middle condition stops *short* of 1 (it says $x < 1$), while the third condition includes it ($x \geq 1$). So 1 belongs to the third branch, and the value the middle branch would have given, $1^2 = 1$, is not $q(1)$. Every input belongs to exactly one branch — that is what makes q a function.

7. Working backwards from an output.

The equation to solve is $|x - 3| + 2 = 6$. Isolate the absolute value first:

$$|x - 3| = 4$$

An absolute value of 4 means the inside is 4 units from zero in *either* direction, so there are two cases:

$$x - 3 = 4 \quad \Rightarrow \quad x = 7 \quad \text{or} \quad x - 3 = -4 \quad \Rightarrow \quad x = -1$$

The two inputs are $\boxed{x = -1}$ and $\boxed{x = 7}$.

Check against the picture: the line $y = 6$ cuts the V once on each arm, at 4 units to the left of the vertex and 4 units to the right. Both answers are the same distance from 3 — that symmetry is a free check on this kind of problem, and a student who reports only $x = 7$ has found one arm and stopped.

8. Using the model twice: forwards and backwards.

(a) Three and a half hours is inside $1 < t \leq 5$, so the middle branch applies:

$$C(3.5) = 4 + 3(3.5 - 1) = 4 + 3(2.5) = 4 + 7.5$$

The cost is $\boxed{11.50}$ dollars.

(b) Now the output is known and the input is not. Set the middle branch equal to the flat rate and solve:

$$4 + 3(t - 1) = 16 \quad \Rightarrow \quad 3(t - 1) = 12 \quad \Rightarrow \quad t - 1 = 4$$

So $t = \boxed{5}$ hours.

Why that number matters: the middle branch reaches 16 at exactly the hour where the third branch takes over. The garage's price ladder is built so the cost does not jump at the changeover — parking longer than five hours simply stops adding cost. A model that did jump there would let a driver pay less by staying longer.

9. Explaining the boundary. (Open response — read the reasoning.)

A full-credit answer makes three points.

1. Why they disagree. $3x + 1$ and $x^2 - 3$ are two different functions. They happen to be glued together into one rule, and there is no reason two different formulas should agree at a shared input. At $x = 2$ they give 7 and 1.

2. Which one wins. The *condition*, not the formula, decides. The top branch carries the condition $x < 2$, which 2 fails; the bottom branch carries $x \geq 2$, which 2 satisfies. So $p(2) = 1$. Every input satisfies exactly one condition, which is what keeps p a function — if both conditions had used $\leq$, the rule would assign two outputs to 2 and would not be a function at all.

3. What the graph does. The line $y = 3x + 1$ approaches the height 7 as x comes up to 2 from the left, but never reaches it (open circle at $(2, 7)$); the parabola starts at $(2, 1)$ with a closed circle. So the graph **jumps** at $x = 2$ — a break of 6 units. The function is discontinuous there.

Accept any answer that names the condition as the decider and describes a break, gap, or jump in the graph. Do not accept “you use the second one because it is written second”.

10. Challenge: where the two branch formulas agree.

Set the two outputs equal — that is what “same output” means — and solve the system

$$y = 3x + 1 \quad \text{and} \quad y = x^2 - 3.$$

Substituting the first into the second:

$$3x + 1 = x^2 - 3 \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x - 4)(x + 1) = 0$$

so $x = 4$ or $x = -1$. Back-substitute into $y = 3x + 1$:

$$x = 4 : y = 13 \quad x = -1 : y = -2$$

The two formulas agree at $\boxed{(4, 13)}$ and at $\boxed{(-1, -2)}$.

Check: $4^2 - 3 = 13 \checkmark \quad (-1)^2 - 3 = -2 \checkmark$

The second half of the question. Neither point makes p continuous, and the reason is the conditions. p only switches branches at $x = 2$, and 2 is not one of these two inputs. At $x = 4$ the rule uses the parabola on both sides; at $x = -1$ it uses the line on both sides. The agreement at these points is real arithmetic, but p never uses it: a piecewise rule is continuous at its switch point only if the two formulas agree *there*, and here they differ by 6 at $x = 2$.

This is the design question behind the parking model in problems 5 and 8, where the formulas were chosen so they *do* agree at the changeover.